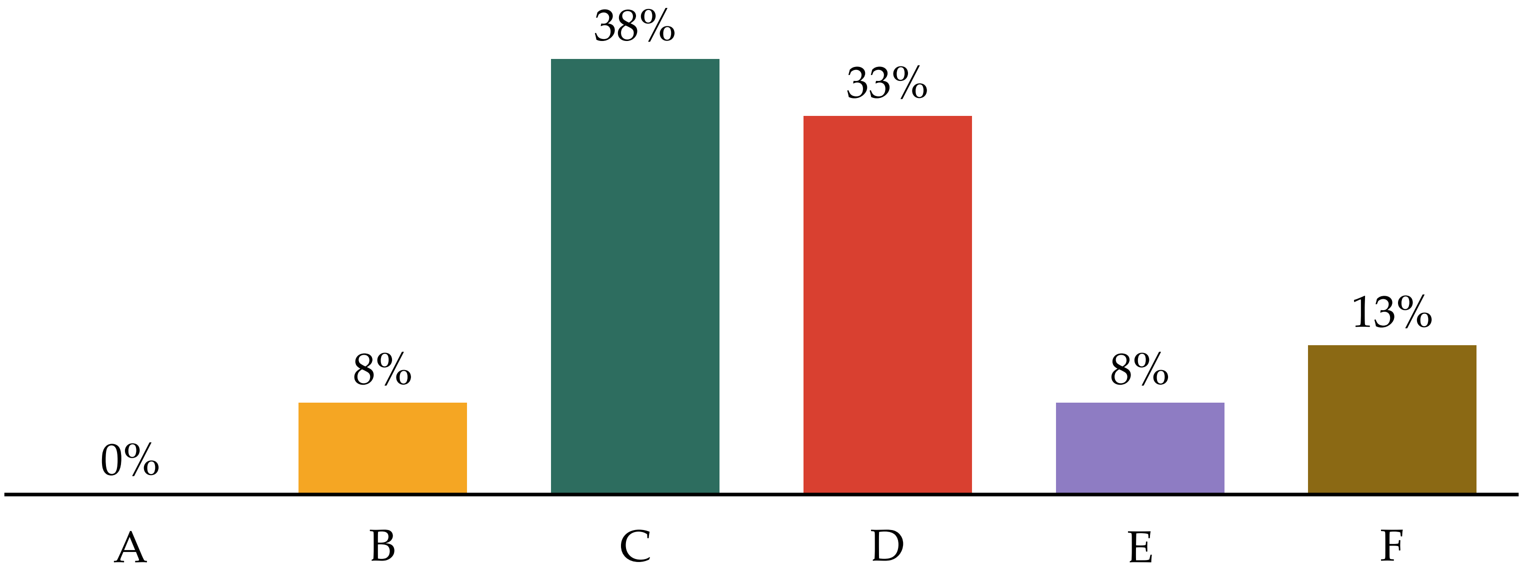
ECON 1550: International Finance

Exchange Rates and the Foreign Exchange Market: An Asset Approach

Announcements

- Problem Set 2 due Wednesday before midnight
- Survey result on speed of lecture
- Office hours and section locations

Survey Results: Speed of Lecture



A	Way too fast
B	A bit too fast
C	Just right
D	A bit too slow
E	Way too slow
F	Overall OK, but fast/slow in the wrong places

Office Hours and Sections

Day / Time	Who	Type	Location
Mon 3:00–4:00 PM	Prof. Duarte	Office hours	Here!
Mon 4:00–6:00 PM	Nathalie	Section/Office hours	Maxcy Hall, Room 010
Tue 4:00–5:00 PM	Raisa	Section	TBD
Wed 12:00–2:00 PM	Eric	Section/Office hours	Salomon 003
Wed 3:00–3:45 PM	Leo	Office hours	Sayles Hall Room 205
Wed 4:00–5:00 PM	Raisa	Office hours	Sayles Hall Room 205
Wed 5:45–7:00 PM	Leo	Office hours	TBD
Fri 12:00–1:00 PM	Prof. Duarte	Office hours	70 Waterman, Room 309

CHECK !!

Agenda

- Finish asset approach to exchange rate determination
  - UIP
  - Carry trade
- Next up: Money market

A model of exchange rate determination

Exogenous variables

Variable	Description
R	Domestic interest rate
R^*	Foreign interest rate
E^e	Expected exchange rate

Endogenous variables

Variable	Description	Equation	Type of equation
E	Exchange rate	$R = R^* + \frac{E^e - E}{E}$	Equilibrium condition

A model of exchange rate determination

- Two investment opportunities
  - Domestic bond with return $R_{\$}$ in Dollars
  - Foreign bond with return R^* in Euros

- To be indifferent between the two investments, the **uncovered interest parity condition** must hold:

$$\text{(UIP): } R_{\$} = R^* + \frac{E^e_{\$/\text{EUR}}}{E_{\$/\text{EUR}}} - 1$$

Exchange Rates

E^e_{t+1} BEST GUESS OF E_{t+1}

t $\qquad\qquad\qquad\qquad$ $t+1$

E_t $\qquad\qquad\qquad\qquad$ E_{t+1}

E_t IS # DOLLARS NEEDED TO BUY 1€ AT t

$E_t \uparrow$ DEPPRECIATION OF $

E_{t+1} NOT KNOWN AT t

Bonds

t $\qquad\qquad\qquad\qquad\qquad\qquad$ $t+1$

P_t $\qquad\qquad\qquad\qquad\qquad\qquad$ \$100

$$R_t = \frac{\$100 - P_t}{P_t}$$

€100

P^*

R^*

Realized vs Expected Returns

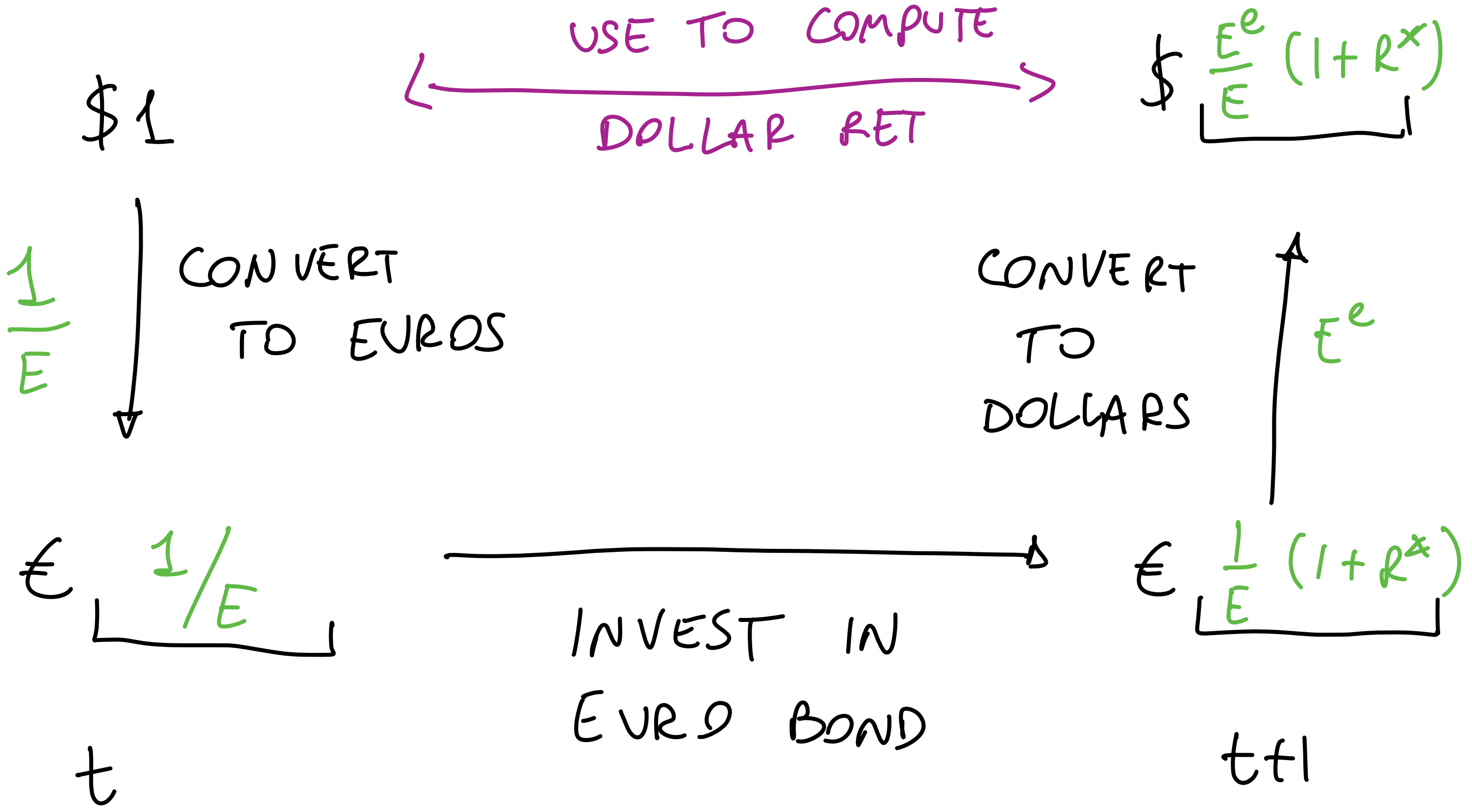
DOLLAR BOND: $\$1 \longrightarrow \$1 \times (1+R)$

$t \qquad\qquad t+1$

EURO BOND: $€1 \longrightarrow €1 \times (1+R^*)$

$t \qquad\qquad t+1$

Expected Returns of Foreign Bond



Approximations

$1+R \rightarrow$ DOLLAR RET ON DOLLAR BOND

$\frac{E^e}{E}(1+R^*) \rightarrow$ DOLLAR RET ON EURO BOND

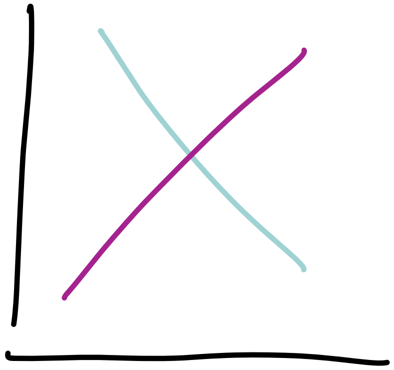
(UIP) $$1+R = \frac{E^e}{E}(1+R^*)$$

Approximations

$$(1+x)(1+y) = 1 + x + y + \underbrace{xy}_{\text{IGNORE!}} \approx 1 + x + y$$

$$\frac{1+x}{1+y} \approx 1 + x - y$$

$$\frac{1+x}{1-y} \approx 1 + x + y$$



LINEAR!

Uncovered Interest Parity

- For both strategies to have the same expected return

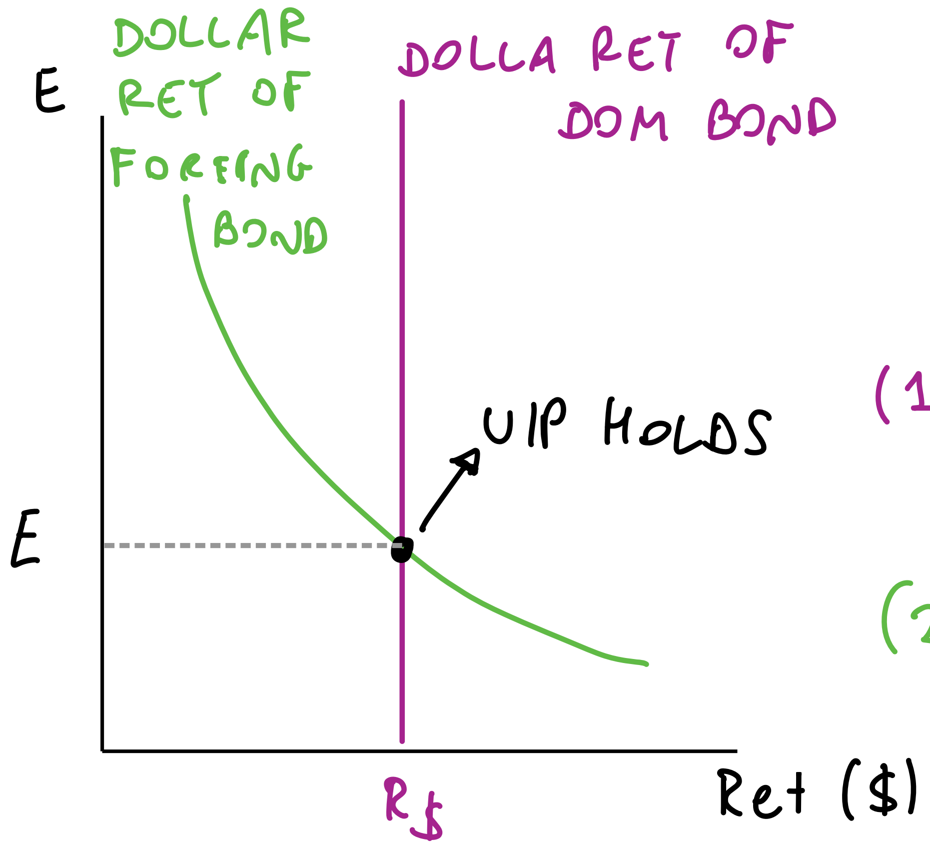
(EXACT UIP): $$1 + R_{\$} = \frac{E^e_{\$/\text{EUR}}}{E_{\$/\text{EUR}}}(1 + R^*)$$

- Approximating

(UIP): $$R_{\$} = R^* + \frac{E^e_{\$/\text{EUR}}}{E_{\$/\text{EUR}}} - 1$$

E ↑ DEPR. DOLLAR WORTH LESS

Equilibrium in Foreign Exchange Market

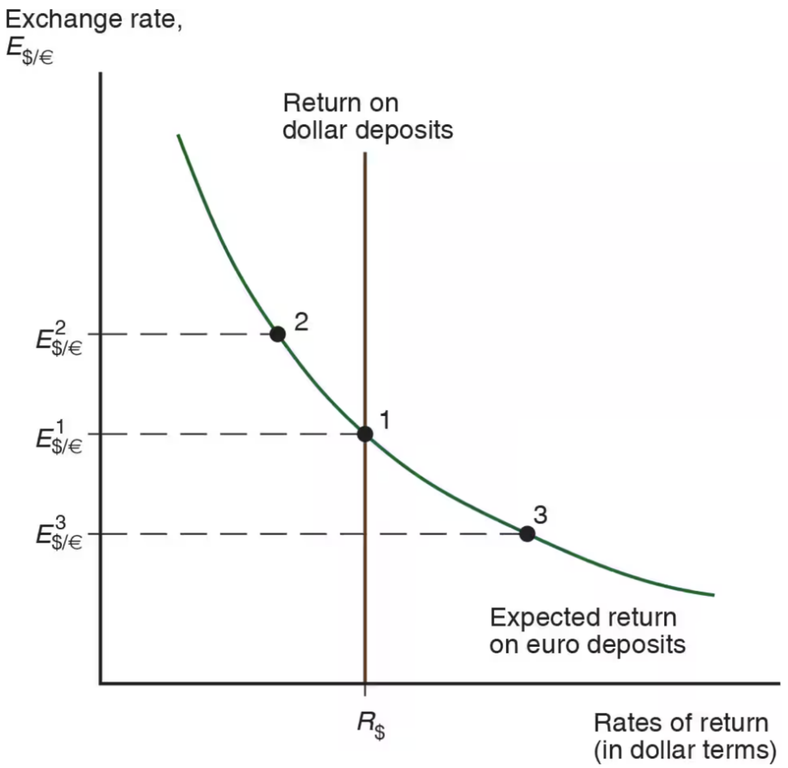


$$\text{UIP: } R_\$ = R^* + \frac{E^e_{\$/\text{EUR}}}{E_{\$/\text{EUR}}} - 1$$

(1) $\text{Ret} = R_\$$

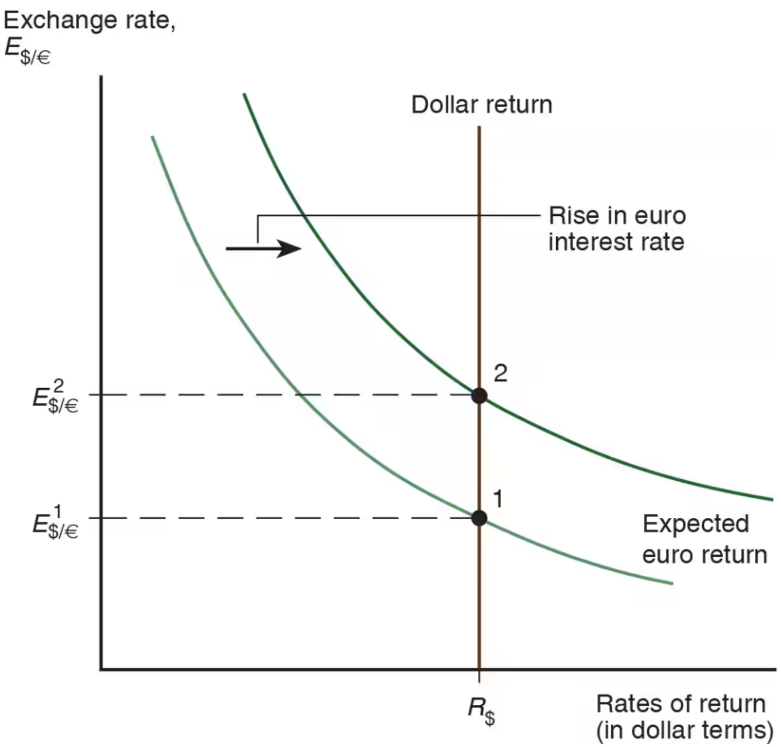
(2) $\text{Ret} = R^* + \frac{E^e}{E} - 1$

Equilibrium in Foreign Exchange Market



$$\text{UIP:} \;\; R_\$ = R^* + \frac{E^e_{\$/\text{EUR}}}{E_{\$/\text{EUR}}} - 1$$

Shocks: Rise in Euro Interest Rate



$R^* \uparrow$

(1) $Ret = R_\$$

(2) $Ret = R^* + \frac{E^e}{E} - 1$

SHIFT OF (2) TO THE RIGHT.

$E \uparrow$ \$ DEPRECIATES!

The carry trade

- Borrowing at "low" rate R and lending at "high" rate R^* is a **carry trade**

$$\begin{array}{c}\text{expected return}\\ \text{on carry trade}\end{array} = R^* + \left(\frac{E^e}{E} - 1\right) - R + \text{risk premium}$$

- Risk: Future exchange rate is not known when we start the carry trade, E^e can be different from the realized future exchange rate

Empirical Failure of UIP

Eight currency portfolios sorted by interest rate differential (portfolio 1 = lowest rates, portfolio 8 = highest).

High interest rate currencies earn higher mean excess returns and have higher Sharpe ratios.

Annual data, 1953–2002, US investor perspective.

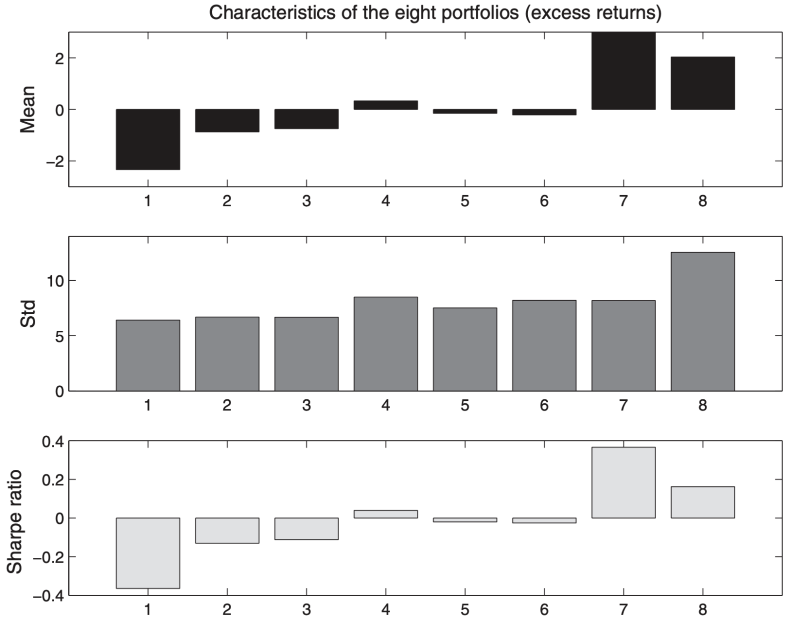


Figure 1. Eight Currency Portfolios

Source: Lustig & Verdelhan (2007), *American Economic Review*

Explanations For Carry Risk Premium

- Crash risk / peso problem
- Correlation with consumption
- Global Volatility Risk
- Liquidity
- Constrained intermediaries

Crash Risk

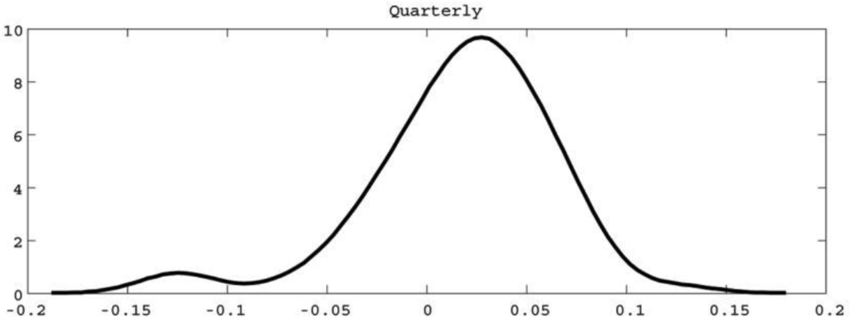


Fig. 1. U.S. dollar/Japanese yen exchange rate from 1996 to 2000

Kernel density of excess returns on a carry trade portfolio (long three high interest currencies, short three low interest currencies). Right: USD/JPY exchange rate, 1996–2000.

Source: Brunnermeier, Nagel, and Pedersen (2008), *NBER Macroeconomics Annual*

Consumption Risk

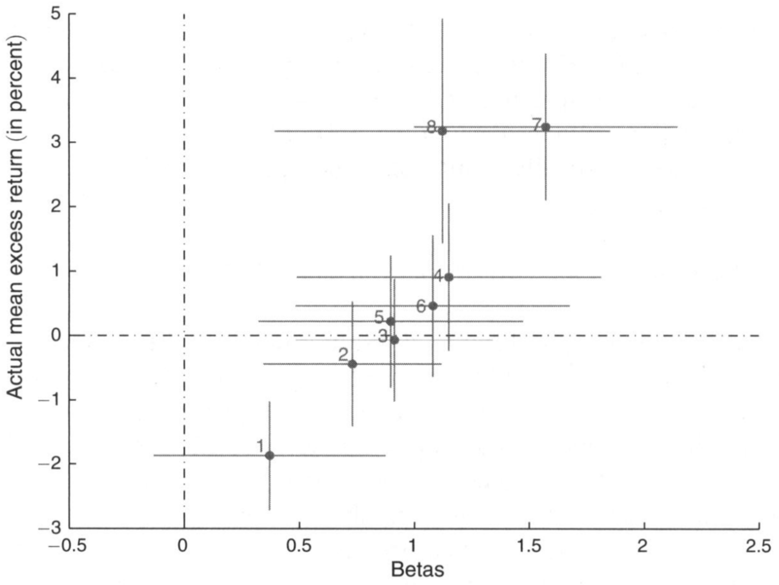


FIGURE 3. DURABLE CONSUMPTION GROWTH BETAS AND AVERAGE CURRENCY EXCESS RETURNS

The dots represent point estimates; the lines represent one standard deviation above and below. The sample is 1953–2008, annual data. Higher consumption betas correspond to higher currency excess returns.

Source: Lustig & Verdelhan (2011), *American Economic Review*

Global Volatility Risk

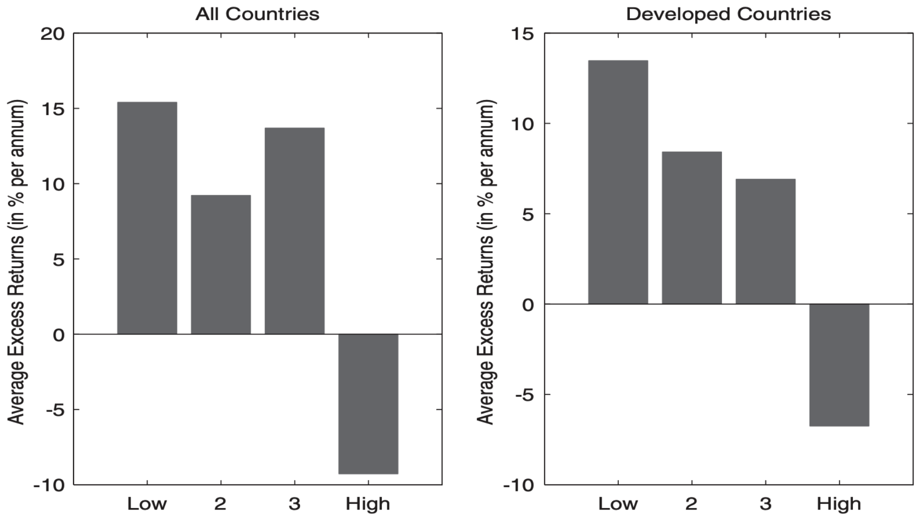
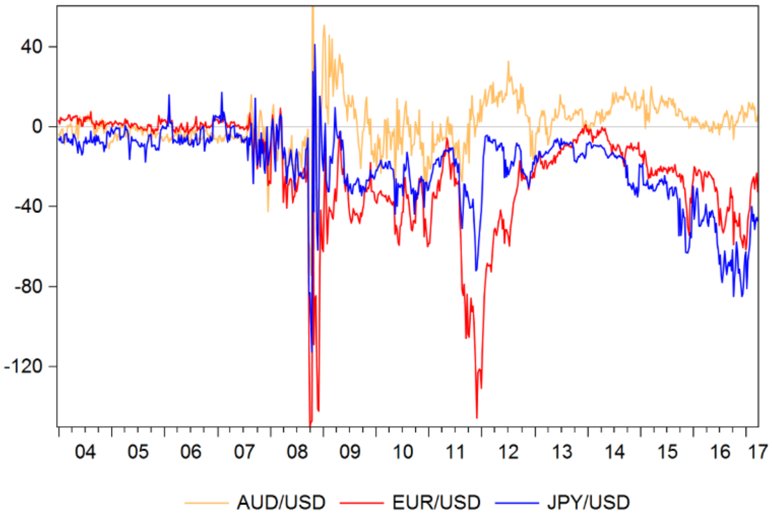


Figure 2. Excess returns and volatility. The figure shows mean excess returns for carry trade portfolios conditional on global FX volatility innovations being within the lowest to highest quartile of its sample distribution (four categories from "Low" to "High" shown on the x-axis of each panel). The bars show average excess returns for being long in portfolio 5 (largest forward discounts) and short in portfolio 1 (lowest forward discounts). The left panel shows results for all countries, while the right panel shows results for developed countries. The sample period is November 1983 to August 2009.

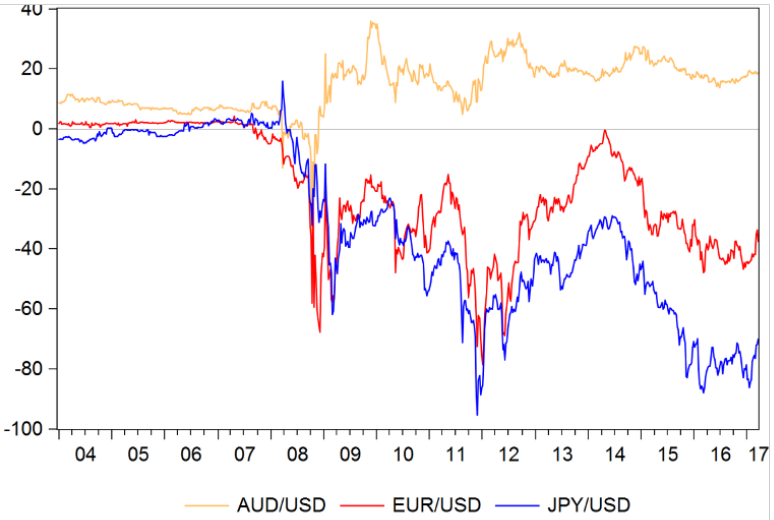
Mean excess returns for carry trade portfolios conditional on global FX volatility innovations being within the lowest to highest quartile. Carry trade earns high returns when volatility is low, but suffers large losses when volatility is high.

Source: Menkhoff, Sarno, Schmeling, and Schrimpf (2012), *Journal of Finance*

Violations of *Covered* Interest Parity



3-month basis: b_{3m}

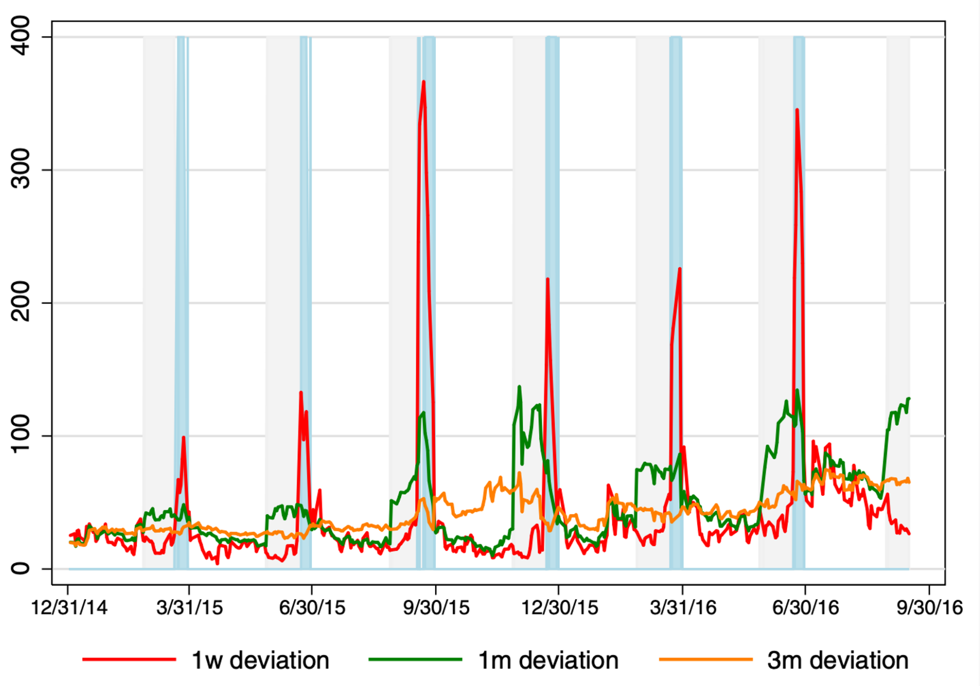


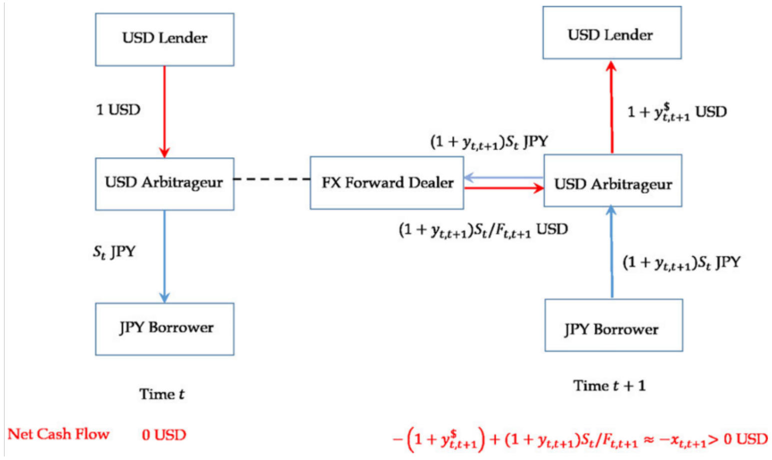
3-year basis: b_{3y}

Source: Sushko, Borio, McCauley, and McGuire, "The Failure of Covered Interest Parity"

Intermediaries are Constrained

Illustration of quarter-end dynamics of CIP deviations.





Source: Du, Tepper, and Verdelhan, "Deviations from Covered Interest Rate Parity," *Journal of Finance*